
Amazon CloudFront Overview

What is Amazon CloudFront?

Amazon CloudFront is a **Content Delivery Network (CDN)** service that securely delivers data, videos, applications, and APIs to users globally with **low latency and high transfer speeds**.

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Key Features

Feature	Description
Global Edge Network	CloudFront uses AWS Edge Locations in 100+ countries to deliver content faster.
Caching	Static and dynamic content is cached closer to users to reduce origin load.
Origin Support	Works with AWS origins (like S3, EC2, ALB) or custom origins (e.g., on-premises servers).
HTTPS & TLS	Secure delivery using SSL/TLS; supports custom SSL certificates.
DDoS Protection	Integrated with AWS Shield Standard for automatic DDoS protection.
Lambda@Edge	Run code closer to users for custom authentication, redirects, header rewriting, etc.
Signed URLs/Cookies	Secure restricted content using token-based access.

Real-Time Metrics Integrated with CloudWatch for monitoring request counts, cache hits, etc.

What are Edge Locations?

- **Edge Locations** are **datacenters** where CloudFront caches content close to the users.
 - These are **not** AWS Regions or Availability Zones.
 - Edge locations are used for:
 - Caching content
 - Running **Lambda@Edge** functions
 - Handling requests/responses
 - Terminating HTTPS
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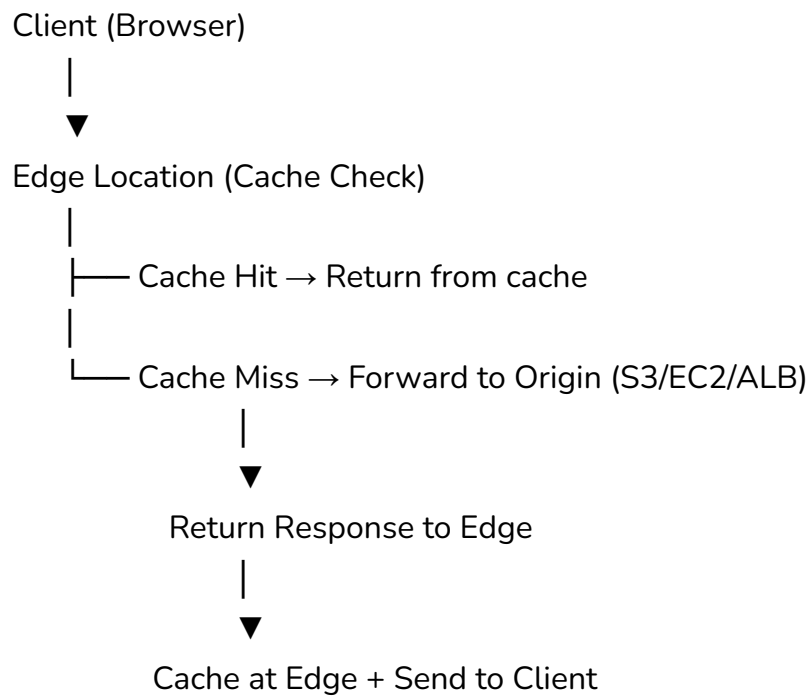
How CloudFront Works (Flow)

1. **User requests content (e.g., image, video, API)** via a CloudFront URL.
2. Request goes to the **nearest Edge Location**.
3. If content is cached:
 -  Delivered directly (cache hit).

4. If content is not cached:

-  CloudFront retrieves it from the **origin** (S3, EC2, etc.), caches it, and then serves it.

CloudFront Architecture Diagram (Text)



Common Use Cases

- Delivering websites and SPAs (Single Page Apps)
- Streaming video (HLS, DASH)
- Distributing software downloads

- Protecting APIs with WAF and rate limiting
 - Reducing S3/EC2 load with caching
 - Restricting access using signed URLs or geo-blocking
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Security & Access

- **Field-Level Encryption**
 - **Geo Restriction** (Allow/Block by country)
 - **Signed URLs / Cookies**
 - **Custom SSL certificates**
 - Integration with **AWS WAF** and **Shield**
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Monitoring & Logs

- Use **CloudWatch** for:
 - Cache hit ratio
 - Latency
 - Request count
- Enable **CloudFront Access Logs** (delivered to S3)

- Use **AWS X-Ray** for tracing requests (with Lambda@Edge)
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Best Practices

- Set **cache TTLs** appropriately
 - Use **compression** (Gzip/Brotli)
 - Serve content over HTTPS
 - Use **origin groups** for failover (e.g., S3 + EC2 backup)
 - Use **Lambda@Edge** for dynamic logic at edge
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Integration Examples

Use Case	CloudFront + Service
Static Website	CloudFront + S3
Secure API	CloudFront + API Gateway
App with Auth	CloudFront + ALB + Cognito
Global Function Execution	CloudFront + Lambda@Edge
